School of Physics,
Engineering and
Computer Science

MSc Data Science Project
7PAM2002-0509-2024
Department of Physics, Astronomy and Mathematics

Data Science FINAL PROJECT REPORT

Project Title:

Forecasting Glucose Levels in Type 1 Diabetes Using Time-Series Approaches

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Date Submitted: **28/08/2025**

Word Count: **4,099 (excluding references and appendices)**

GitHub Repository: https://github.com/saeedsahar/GlucoseForecastingML.git

DECLARATION STATEMENT

This report is submitted in partial fulfilment of the requirement for the degree of Master of Science **in Data Science** at the University of Hertfordshire.

I have read the detailed guidance to students on academic integrity, misconduct and plagiarism information at Assessment Offences and Academic Misconduct and understand the University process of dealing with suspected cases of academic misconduct and the possible penalties, which could include failing the project or course.

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I did not use human participants in my MSc Project.

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Acknowledgement

I would like to sincerely thank my supervisor, **Dr. William Cooper**, for his invaluable guidance, encouragement, and constructive feedback throughout this project. His expertise and support have been instrumental in shaping my research and helping me stay focused.

I am also grateful to the **University of Hertfordshire's MSc Data Science programme team** for providing the resources and academic environment that made this work possible.

Finally, I would like to thank my **family and friends** for their patience, understanding, and constant motivation during this journey. Their support has been my greatest source of strength.

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Abstract

Accurate forecasting of blood glucose levels is a vital part of diabetes management, helping to prevent sudden spikes or drops and supporting more proactive care. This study applies machine learning and deep learning techniques to predict short-term glucose trends using a rich dataset of continuous glucose monitoring (CGM) records combined with demographic, lifestyle, and clinical information. Through careful feature engineering and exploratory data analysis, we prepare the data and evaluate three predictive models: Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and Convolutional Neural Network (CNN). By comparing their performance, the study highlights the strengths and limitations of each approach. The results show that CNN consistently delivered the most accurate and stable forecasts, especially during sharp glucose fluctuations. These findings demonstrate the potential of data driven forecasting to support earlier interventions and improve the day to day management of diabetes.

1. Introduction

Diabetes is a lifelong condition that needs constant attention to avoid sudden swings in blood sugar. Sharp increases (hyperglycemia) or drops (hypoglycemia) can be dangerous, so being able to predict these changes before they happen is a big step in keeping patients safe (American Diabetes Association, 2022)**.**

With the help of Continuous Glucose Monitoring (CGM) devices, we now have a steady stream of real-time data that opens up new possibilities for prediction (Bequette, 2019) Using this data, advanced models can move beyond simple tracking and actually start forecasting, giving people and clinicians a chance to act before problems arise.

This project looks at how deep learning can help with that challenge. In particular, it compares three different models—LSTM, GRU, and CNN. LSTM and GRU are designed to recognise patterns over time, while CNN, better known for image recognition, has also shown potential in handling time-series data (Zhu, et al., 2020)**.**

By putting these models head-to-head on the same dataset, the study aims to give a clear picture of their strengths and weaknesses. Ultimately, the goal is to see how such predictive tools could support day-to-day diabetes care, helping to prevent complications and make life easier for patients.

1.1 Research Questions

This project explores how well we can predict future glucose levels using continuous glucose monitoring (CGM) data. The main questions are:

1. **Which model works best?**
   Out of LSTM, GRU, and CNN, which gives the most accurate glucose predictions?
2. **Does tuning help?**
   If we adjust settings like units, filters, dropout, and learning rate, do the models actually get better?
3. **Can the models work for different people?**
   Since every patient is different, can the models still predict well for new patients they haven't seen before?
4. **Are the predictions useful in real life?**
   Can the models spot important events, like high glucose levels (above 140 mg/dL), in a way that would help with daily diabetes management?

2. Literature Review

Forecasting glucose levels has become a fast-growing research area, driven by the increasing availability of data from Continuous Glucose Monitoring (CGM) devices. Early studies often relied on traditional statistical methods like ARIMA, but these models struggled with the complex and highly non-linear behaviour of glucose in the body (Zhang, et al., 2019)

With the rise of deep learning, researchers began turning to recurrent neural networks. Long Short-Term Memory (LSTM) networks have quickly become a popular choice because of their ability to capture long-term dependencies in time-series data (Zhao, et al., 2021)Gated Recurrent Units (GRUs) followed as a lighter, faster version of LSTMs, often providing similar levels of accuracy with fewer parameters (Li, et al., 2019)Both approaches have shown strong results in predicting short-term fluctuations in blood glucose.

More recently, Convolutional Neural Networks (CNNs) have been applied to sequential biomedical data. Although CNNs were originally built for image recognition, their strength in detecting local patterns has proven useful in time-series forecasting as well. Several studies report that CNN-based models can even outperform recurrent models, offering both higher accuracy and faster training times (Sun, et al., 2021).

Taken together, the literature shows that LSTMs and GRUs remain reliable baselines, but CNNs are emerging as serious contenders. Still, most comparative studies remain limited, and performance can vary greatly depending on the dataset and how it is preprocessed (Xie, et al., 2020)

Despite these advances, challenges remain. Missing data, sparse signals, and the "black-box" nature of deep learning models all limit their direct use in clinical settings. For adoption in practice, models must not only be accurate but also interpretable, so that healthcare providers can trust the recommendations they produce (Zuo, et al., 2022)

3. Data Description & Preprocessing

he dataset for this project comes from a continuous glucose monitoring (CGM) study and contains more than 645,000 records across 41 variables, originally made available through **GlucoBench (IrinaStatsLab, 2025)** at **https://github.com/IrinaStatsLab/GlucoBench/.** Each entry represents a patient reading at a given point in time, linked through a unique patient ID. The primary variable is *gl*, which measures glucose levels at each timestamp, but the dataset also provides a wide range of supporting information:

- **Demographics** – patient background details such as gender, race, marital status, education level, and annual income.
- **Lifestyle factors** – records of behaviours including how often patients exercise, how often they drink alcohol during the week, and whether they engage in binge drinking episodes.
- **Clinical history** – variables such as age at Type 1 diabetes diagnosis, number of hospitalisations for diabetic ketoacidosis (DKA), and the frequency of severe hypoglycemia episodes.
- **Comorbidities** – conditions commonly associated with diabetes, including hypertension, kidney disease, and depression.
- **Medication use** – records of treatments ranging from insulin to statins, vitamins, and other prescribed or supplementary medications.

Together, this mix of clinical, behavioural, and demographic data makes the dataset well-suited for investigating how glucose levels are shaped by both short-term behaviour and long-term health conditions (Albers, et al., 2014)

To prepare the dataset for model training, several preprocessing steps were carried out:

- **Data cleaning** – inconsistent or missing glucose readings were identified and removed.
- **Time alignment** – records were ordered chronologically per patient, and irregular intervals were resampled to fixed time steps to keep the data uniform.
- **Scaling** – continuous variables such as glucose, insulin dosage, weight, and income were standardised using z-scores, ensuring comparability across patients.

- **Sequence generation** – overlapping windows of past glucose readings (12 steps, or roughly one hour of history) were created to predict the next **value** (Zhu, et al., 2020)
- **Splitting** – data was divided chronologically into training, validation, and testing sets to prevent leakage and to mirror real-world forecasting conditions.

These steps ensured the dataset was consistent, numerically stable, and properly structured for use with the three deep learning models—LSTM, GRU, and CNN.

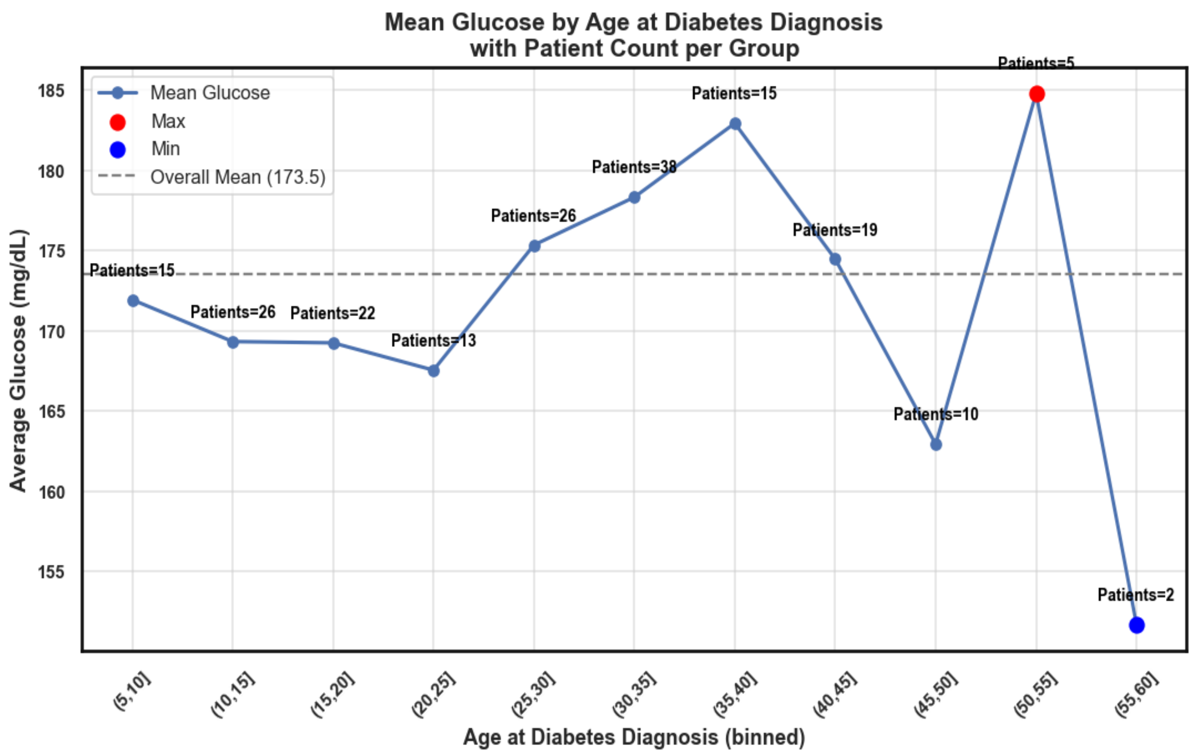
4. Exploratory Data Analysis (EDA)

4.1 Overview

This project examines how social context, lifestyle behaviors, clinical comorbidities, and medications relate to blood glucose in an adult diabetes cohort and prepares the ground for forecasting models.

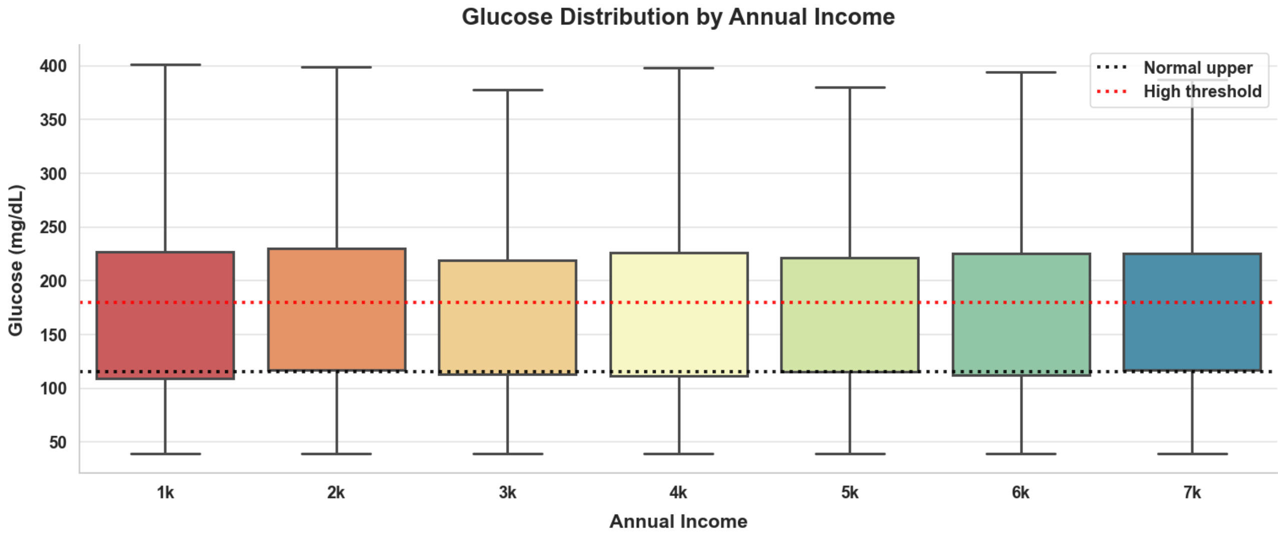
4.2 Socio-demographic factors

4.2.1 Mean Glucose by Age at Diabetes Diagnosis with Patient Counts



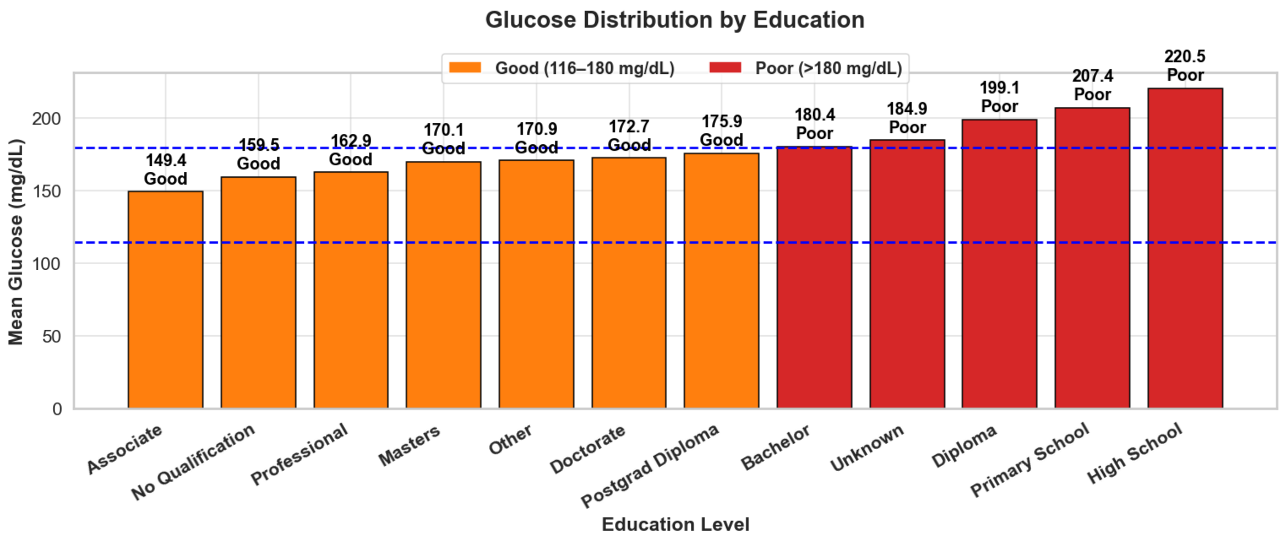
People diagnosed at younger or middle ages show similar average glucose, while some older groups have slightly higher values. Each age group includes many patients, so the trend looks consistent overall.

4.2.2 Income impact on glucose



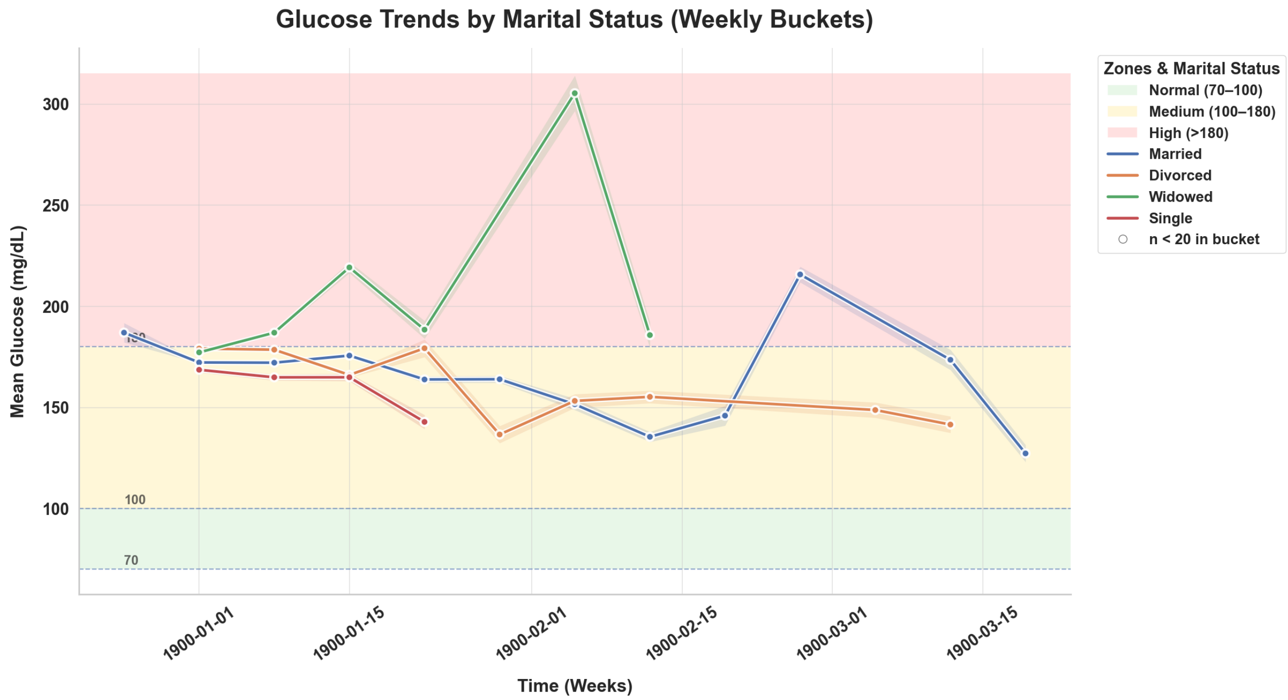
The impact of income on glucose is minimal. Across different income groups, glucose levels show wide variation but no clear upward or downward trend. The statistical correlation is very weak (≈0.015), confirming that income does not meaningfully influence glucose control in this dataset.

4.2.3 Relationship Between Education and Glucose Levels



Glucose levels show a clear link with education. People with higher education (Associate, Professional, Masters, Doctorate) tend to have better glucose control (mostly in the "Good" range), while those with lower education (High School, Primary, Diploma) show poorer control, with mean glucose often above 180 mg/dL. (American Diabetes Association, 2022).

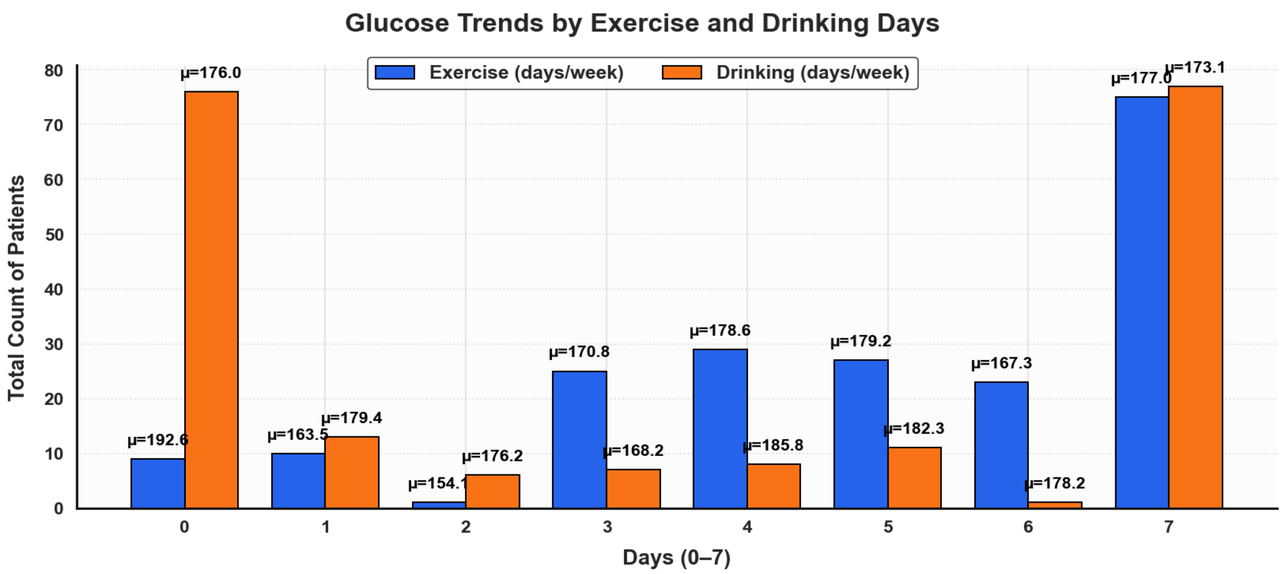
4.2.4 Marital Status and Glucose Trends



The weekly trends show that glucose levels vary across marital groups, but the patterns are uneven. Married individuals tend to display larger swings, occasionally spiking into the high zone (>180 mg/dL). Widowed patients also show instability, with some sharp peaks. In contrast, divorced and single groups remain more steady, usually within the medium range (100–180 mg/dL). Overall, marital status appears to have some effect on glucose control, but the influence is inconsistent and not dominant.

4.3 Lifestyle

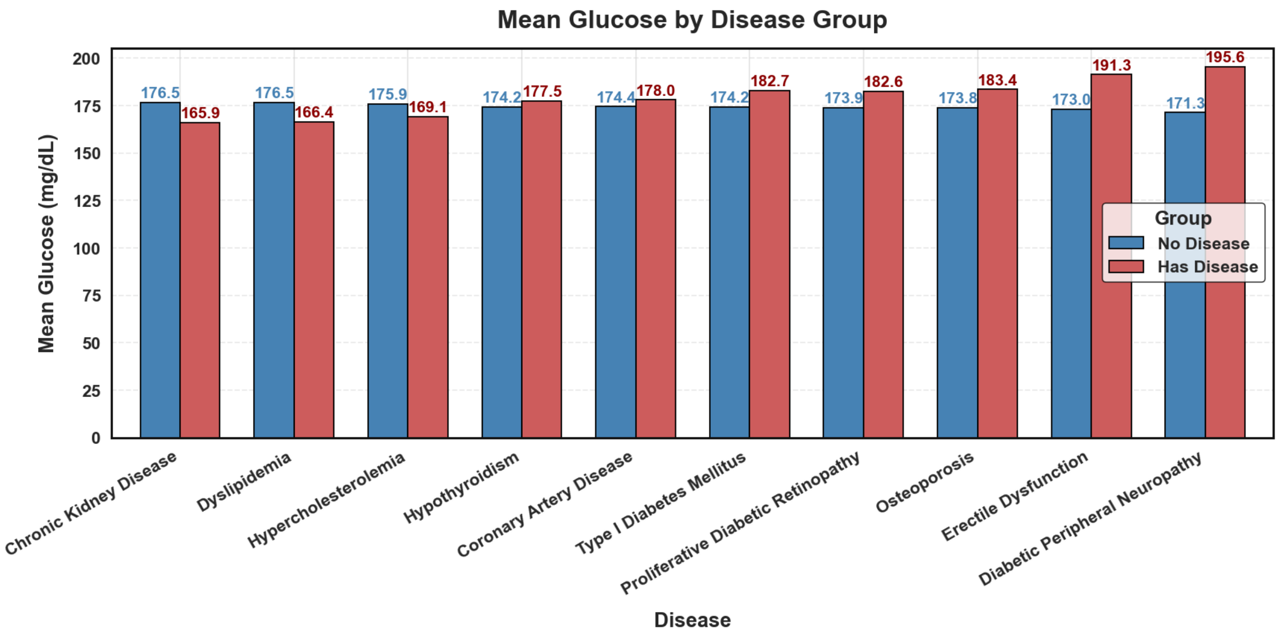
4.3.1 Exercise and Binge Drinking per Week



People who exercise more often usually have lower and steadier glucose levels compared to those who don’t exercise.

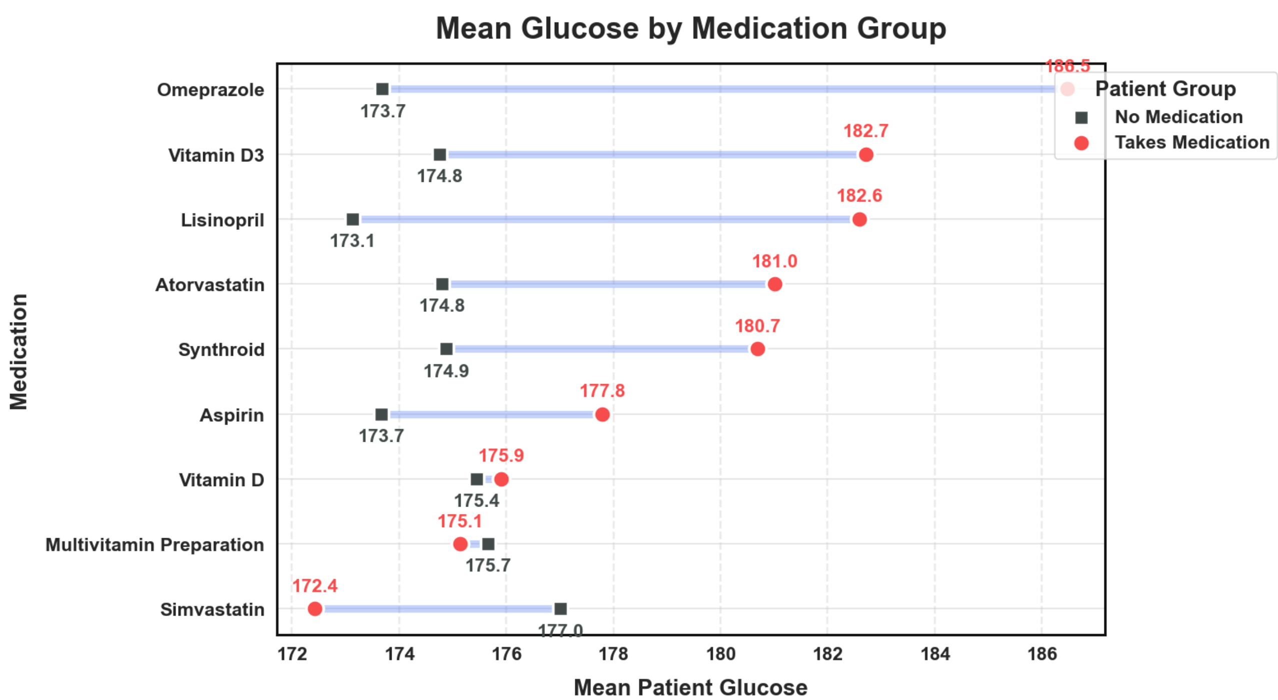
For alcohol drinking, the pattern isn't very clear. Glucose stays in a similar range no matter how many days people drink.

4.3.2 Clinical Comorbidities and Their Impact



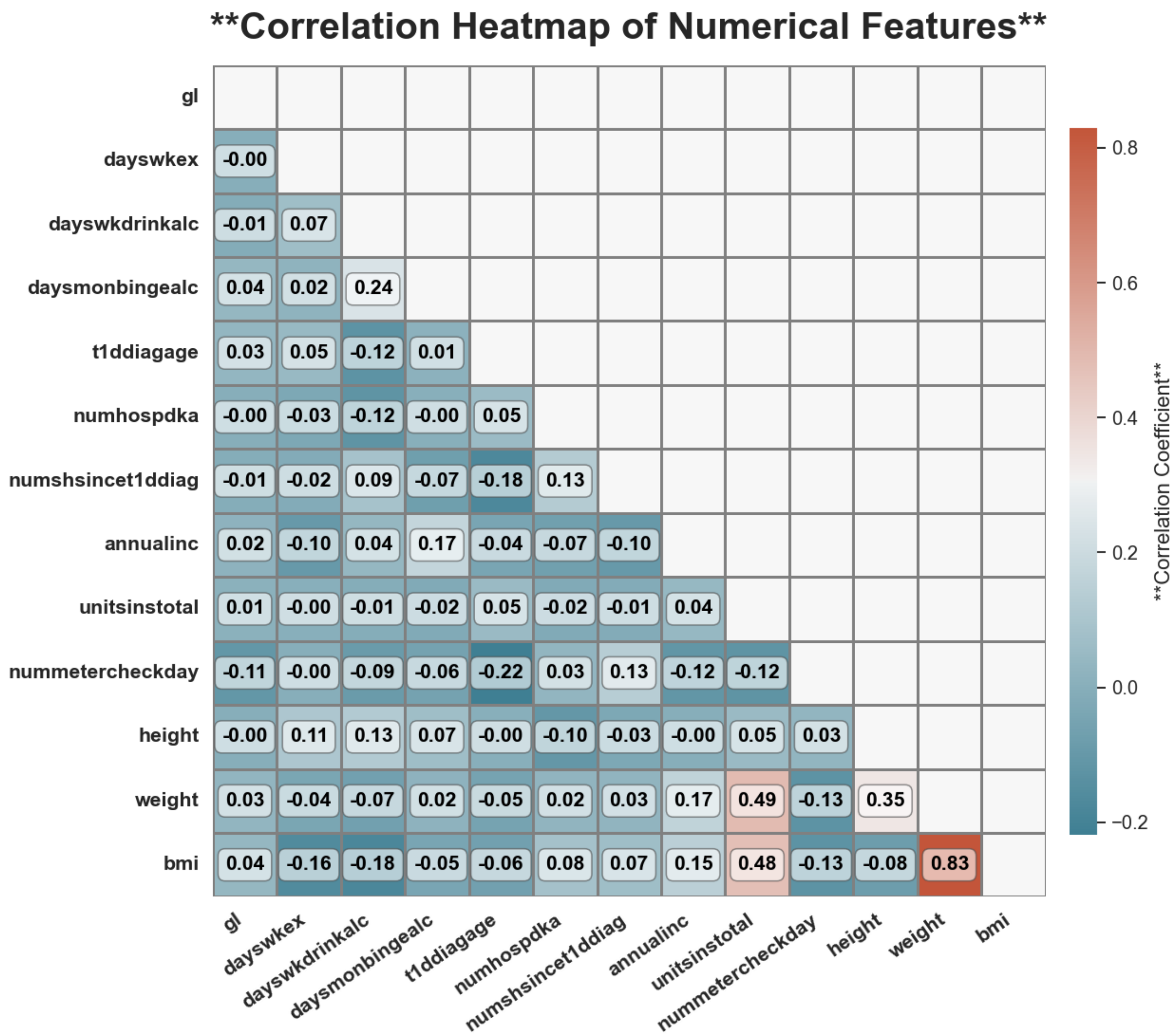
We compared per-patient mean glucose between those with and without each condition (CKD, dyslipidemia, neuropathy, retinopathy, hypothyroidism, etc.) using grouped bar plots. In nearly all cases, individuals with comorbidities show slightly higher glucose levels than those without. The differences are not extreme but consistent, suggesting that added disease burden makes glucose control more difficult. The clearest gaps appear in conditions such as diabetic neuropathy, erectile dysfunction, and retinopathy, where mean glucose is notably elevated (Zuo, 2022).

4.4 Medication and Supplement Usage



While medications such as statins, aspirin, vitamin D, and various supplements showed relatively minor but statistically significant differences in mean insulin usage, these variables add context to each patient's overall care complexity and may be useful as covariates in predictive modeling.

4.5 Correlation Analysis



No strong linear links with **glucose** all |r| are tiny. Lifestyle (exercise/drinking), income, monitoring, etc., show **near-zero** correlation with gl. The only big relationship in the matrix is **BMI and weight** (highly positive), which is expected. This does't rule out **non-linear or lagged** effects; it just says simple linear ties are weak.

4.5.1 Correlation Table with Strength

Variable	Correlation with Target	Strength
id	0.023	Very Weak
gl	1.000	Perfect
dayswkex	-0.004	Very Weak
dayswkdrinkalc	-0.021	Very Weak
daysmonbingealc	0.038	Very Weak
t1ddiagage	0.027	Very Weak
numhospdka	-0.011	Very Weak
numshsincet1ddiag	0.007	Very Weak
annualinc	0.015	Very Weak
unitsinstotal	0.023	Very Weak
nummetercheckday	-0.108	Very Weak
height	-0.002	Very Weak
weight	0.031	Very Weak
bmi	0.037	Very Weak

The only strong link is with gl, since it's the target itself. All other variables show very weak relationships. The small positive ones include bmi, weight, and daysmonbingealc, while the weakest negative link comes from nummetercheckday. Overall, no other column adds much predictive power.

5. Model Development & Methods

5.1 Overview of Predictive Modeling Approach

This project applies **three deep learning models** — LSTM, GRU, and CNN — to predict short-term glucose levels from continuous glucose monitoring (CGM) data. The approach was designed to be simple, logical, and easy to follow:

- **Preprocessing:** Patient data was ordered chronologically, resampled into 5-minute intervals, and cleaned to handle missing or noisy values.
- **Feature Engineering:** Twelve lag features (≈1 hour of glucose history) were created to frame the task as predicting the next reading from past values.
- **Scaling:** StandardScaler was applied to both inputs and outputs, fitted only on training data to avoid leakage; predictions were inverse scaled for interpretation.
- **Data Splitting:** A patient-wise chronological split was used — the last 60 readings reserved for testing, with the rest divided into training (≈70–75%) and validation (≈10–15%).
- **Baseline Models:** Standard versions of LSTM, GRU, and CNN were trained to establish initial performance benchmarks.
- **Hyperparameter Tuning:** Each model explored 12 configurations (varying units, filters, dropout, dense layers, learning rate), with validation RMSE guiding the best setup.
- **Training Setup:** All models used MSE loss, the Adam optimizer, early stopping to prevent overfitting, and consistent batch size/epochs for fairness.
- **Evaluation:** Performance was assessed with regression metrics (R^2, RMSE, MAE) and classification metrics (Accuracy, Precision, Recall, F1, Confusion Matrix at 140 mg/dL).
- **Results & Selection:** Comparisons through tables, scatter plots, training curves, and patient-level studies showed that the tuned CNN achieved the highest R^2 (0.996), lowest RMSE (≈5 mg/dL), and the strongest classification F1-score (0.986), making it the most reliable model overall.

5.2 Feature Engineering

To convert CGM data into supervised sequences suitable for deep learning:

- **Re sampling:** Glucose readings were resampled to fixed **5-minute intervals** to create consistent time steps across all patients.
- **Lag Features:** For each target value (gl), the preceding **12 lags** (lag_1 … lag_12) were used as features. This represents approximately 1 hour of patient glucose history.
- **Windowed Dataset:** Inputs X = {lag_1 … lag_12} and output y = gl(t) were created for each timepoint.
- **Patient-wise Construction:** Features were generated independently for each patient to prevent inter-patient leakage.

This ensured that models learned both short-term temporal dependencies and personalised patterns within patient trajectories.

5.3 Scaling

To keep training stable and results meaningful, we scaled both inputs and outputs with StandardScaler. The scaler was fitted only on the training data (to avoid leakage) and then applied consistently to validation and test sets. This kept features on a comparable scale, sped up convergence, and prevented models from being biased by extreme glucose ranges (50–400 mg/dL).

Predictions were inverse transformed back into mg/dL so metrics like RMSE and MAE stayed clinically interpretable. In short, scaling acted as a training stabiliser and a bridge back to clinical reality.

5.4 Training, Validation, and Testing Split

To keep the setup close to real-life forecasting, we split each patient's data by time rather than at random.

- **Testing:** The latest TEST_WINDOW readings were held back as the "future" the model had to predict.
- **Validation:** From the remaining pool, about 10% of the earliest samples were used to tune the model.
- **Training:** Everything before that went into training.
- This way, the model always learned from past data to predict future trends — just like it would in a clinical setting. Every patient contributed to all three phases, but their most recent readings stayed unseen until the very end. In practice, this worked out to around **70–75% training, 10–15% validation, and 15–20% testing**, a balance that gave the model enough history to learn from while keeping evaluation fair and realistic.

5.4 Model Architectures and Working

5.4.1 Long Short-Term Memory (LSTM)

The Long Short-Term Memory (LSTM) network is a specialised type of recurrent neural network (RNN) built to capture long-term patterns in sequential data. Unlike standard RNNs, which often forget earlier information as sequences grow longer, LSTMs use memory cells and a system of gates (input, forget, and output) to decide what information to keep and what to discard. This design makes them especially effective for modelling glucose levels, where past fluctuations can strongly influence future values (Zhu, et al., 2018).

Working Example (code):

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Dropout
from tensorflow.keras.optimizers import Adam

# LSTM Model
model_lstm = Sequential([
    LSTM(128, input_shape=(12, 1)),    # 12 lags ~ 1 hour history
    Dropout(0.1),                      # prevents overfitting
    Dense(64, activation='relu'),
    Dense(1)                           # output: next glucose value
])
model_lstm.compile(optimizer=Adam(learning_rate=0.0005), loss='mse')
```

This setup allows the model to remember patterns of glucose fluctuations over short and long sequences

5.4.2 Gated Recurrent Unit (GRU)

The Gated Recurrent Unit (GRU) is a streamlined version of the LSTM. Instead of using separate forget and input gates, GRUs combine them into a single update gate, which makes the network simpler and faster to train. Despite this reduced complexity, GRUs are still able to capture sequential patterns effectively and often achieve performance levels close to LSTMs. Their faster training speed makes them particularly useful for clinical time-series tasks such as CGM data, where both accuracy and efficiency matter (Li, et al., 2019).

Working Example (code):

```
from tensorflow.keras.layers import GRU

# GRU Model
model_gru = Sequential([
    GRU(64, input_shape=(12, 1)),      # fewer parameters than LSTM
    Dropout(0.1),
    Dense(32, activation='relu'),
    Dense(1)
])
model_gru.compile(optimizer=Adam(learning_rate=0.001), loss='mse')
```

GRU trades a little accuracy for speed and efficiency, making it attractive in real-time glucose monitoring systems.

5.4.3 Convolutional Neural Network (CNN)

Convolutional Neural Networks (CNNs) are best known for image recognition, but they are also effective for time-series tasks. In this context, CNNs act as pattern detectors: convolutional filters slide across glucose sequences, identifying local trends such as sudden spikes or drops. This ability to capture short-term variations makes CNNs particularly strong for glucose forecasting. Another advantage is that CNNs train faster than LSTMs or GRUs, since they don't depend on sequential recurrence. As a result, they offer both accuracy and efficiency for CGM-based prediction (Sun, et al., 2021).

Working Example (code):

```
from tensorflow.keras.layers import Conv1D, Flatten

# CNN Model
model_cnn = Sequential([
    Conv1D(filters=96, kernel_size=7, activation='relu', input_shape=(12, 1)),
    Flatten(),
    Dropout(0.1),
    Dense(32, activation='relu'),
    Dense(1)
])
model_cnn.compile(optimizer=Adam(learning_rate=0.0005), loss='mse')
```

CNN extracts local "shapes" of glucose movement — peaks, dips, steady lines — making it particularly strong at short-term prediction.

5.5 Baseline Model Performance

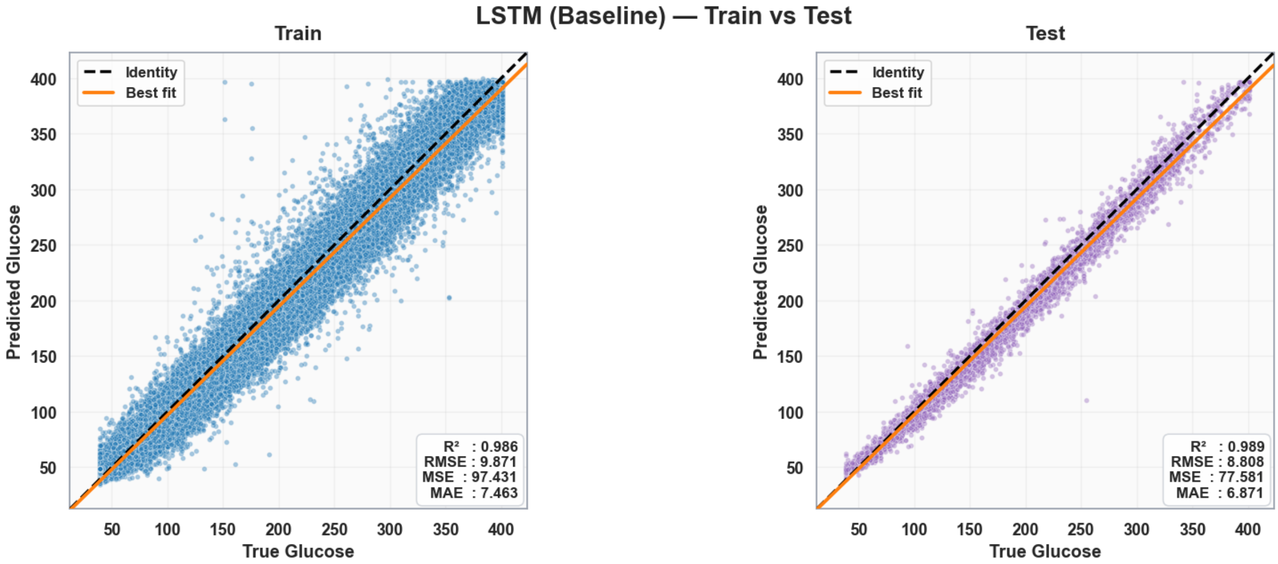
Baseline models (GRU, CNN, and LSTM) were trained with standard configurations to establish reference performance before tuning. Results are summarised in both tabular and visual form for clarity. Baseline models were trained with standard configurations.

Table 5.5.1 Baseline performance across Train/Validation/Test sets

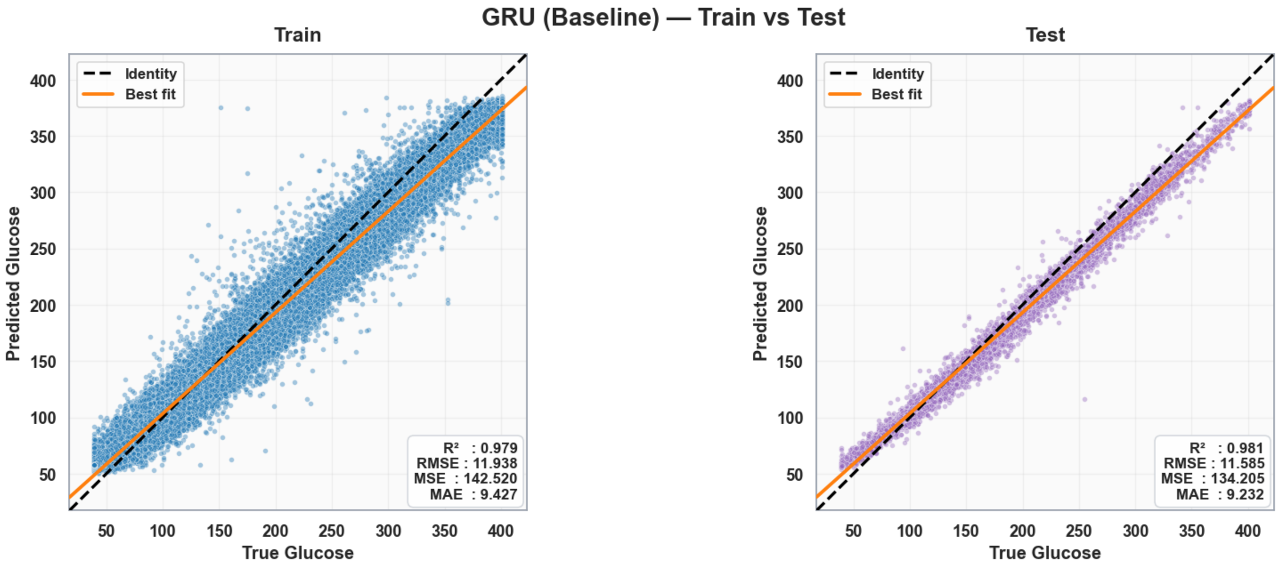
Model	Train R^2	Train RMSE	Train MAE	Val R^2	Val RMSE	Val MAE	Test R^2	Test RMSE	Test MAE
LSTM (B)	0.982	11.013	7.583	0.984	10.692	7.325	0.988	9.160	6.344
GRU (B)	0.979	11.956	9.943	0.980	11.880	9.886	0.981	11.450	9.596
CNN (B)	0.994	6.492	4.390	0.994	6.520	4.330	0.995	5.641	3.783

5.5.2 Baseline Train vs Test Scatter Plots

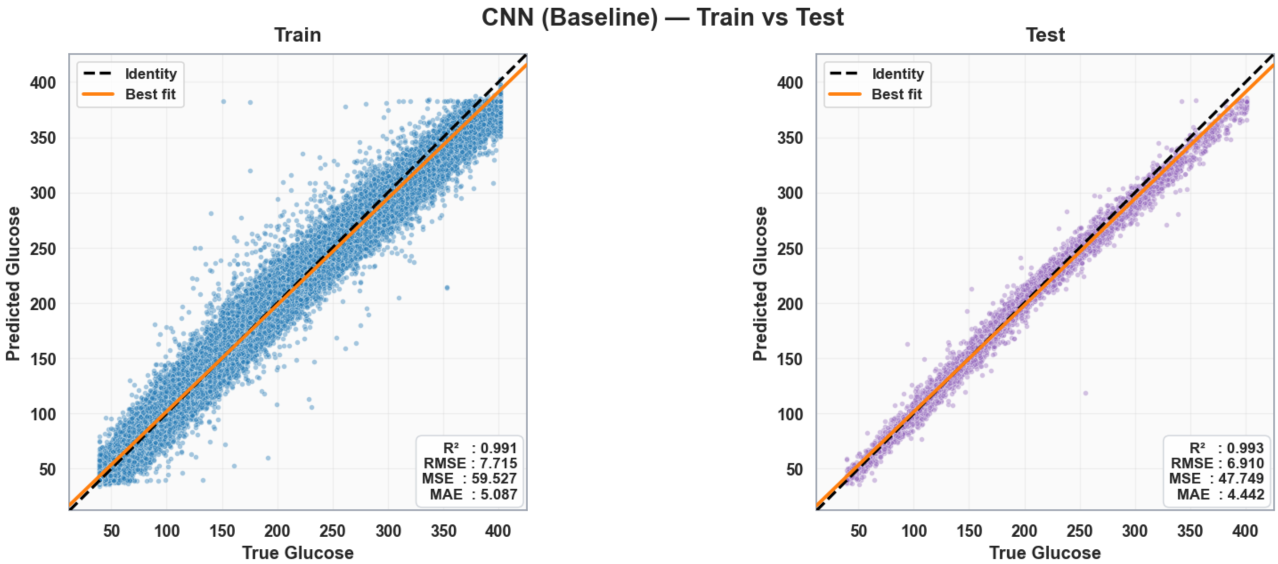
- **LSTM Baseline — Train vs Test**



- **GRU Baseline — Train vs Test**



- **CNN Baseline — Train vs Test**

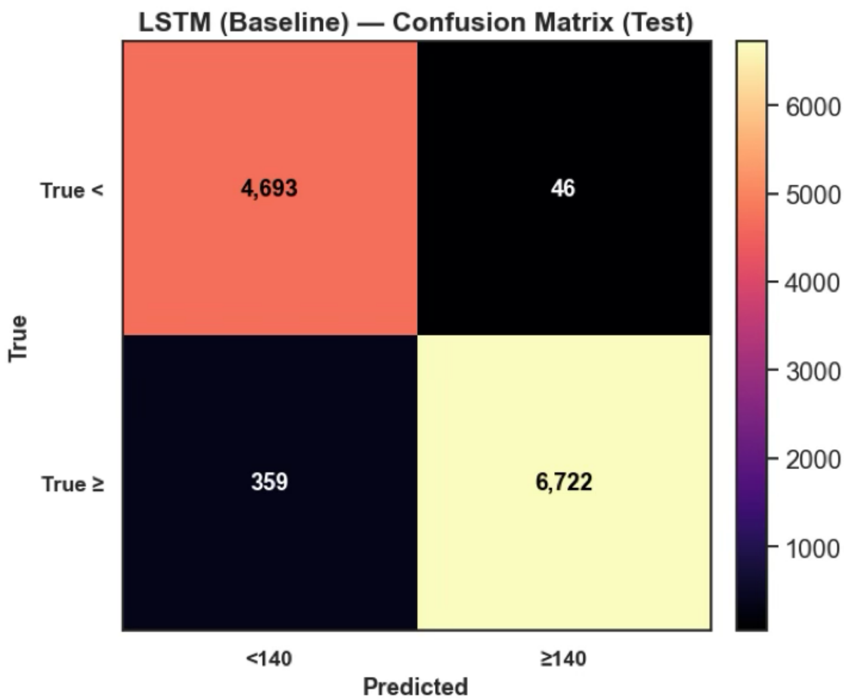


5.5.3 Observations from Baseline Scatter Plots

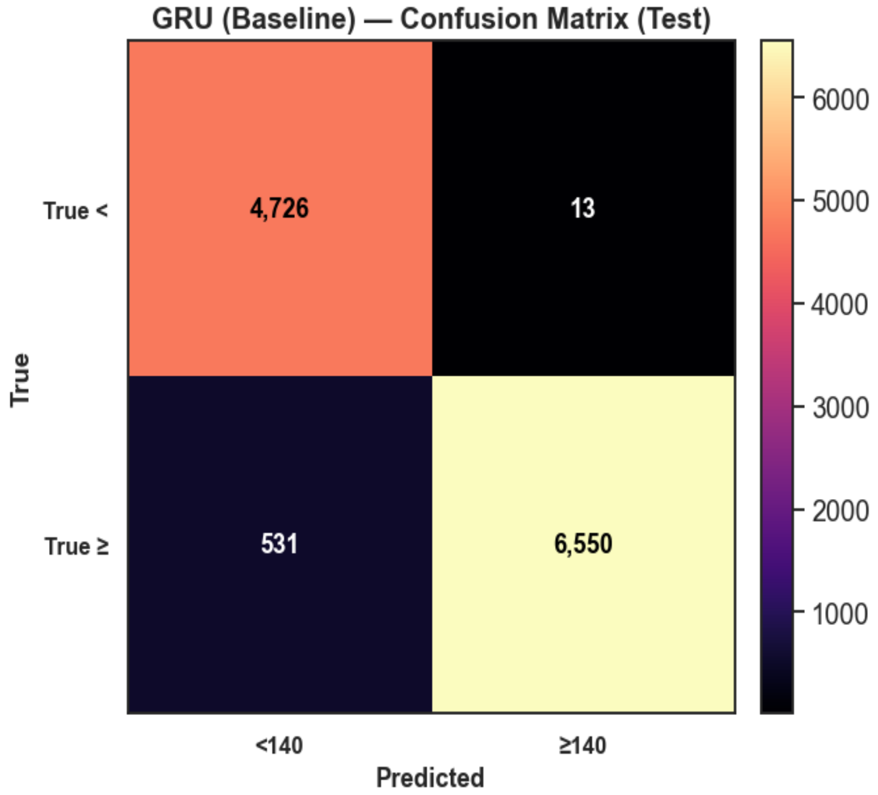
The scatter plots show that all three models capture the glucose trajectory well, with predictions closely following the identity line. GRU performs reliably but shows a bit more spread at higher glucose values, leading to slightly larger errors. CNN demonstrates the tightest fit, with predictions almost perfectly aligned and very low error, making it the strongest baseline. LSTM sits in between: it generalises well with no signs of overfitting, but its errors are somewhat higher than CNN's. Overall, the results highlight that while all models generalise effectively, CNN stands out as the most accurate and stable baseline.

5.5.4 Baseline Confusion Matrices

- **LSTM Baseline — Test**



- **GRU Baseline — Test**



- **CNN Baseline — Test**

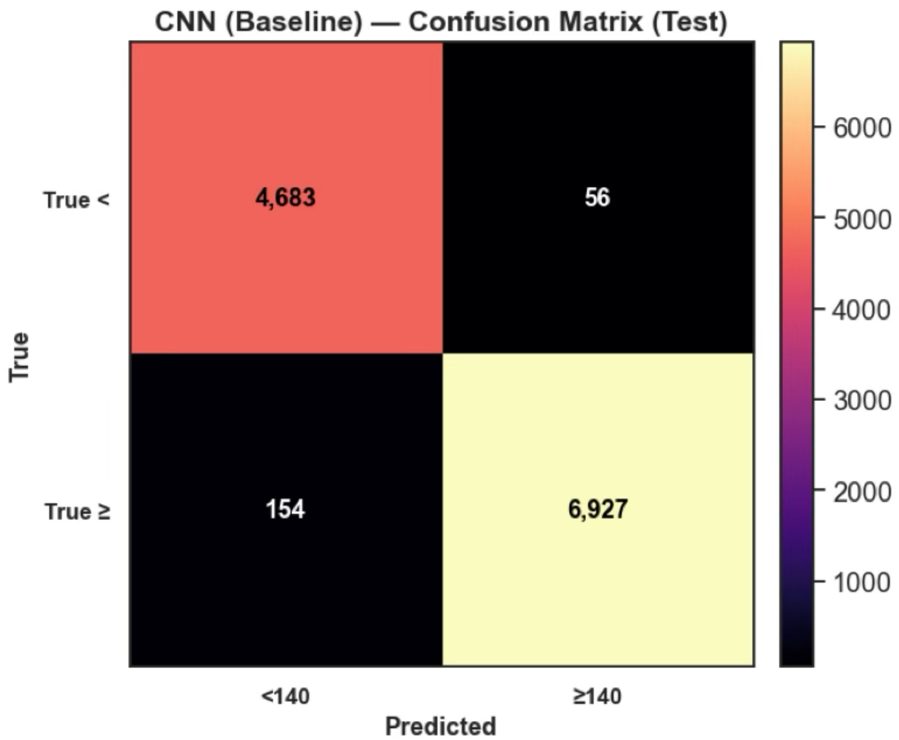


Table 5.5.5 Confusion Matrix Results and Metrics for Baseline Models (Threshold = 140 mg/dL)

Model	TN	FP	FN	TP	Accuracy	Sensitivity (≥140)	Specificity (<140)
CNN (Baseline)	4683	56	154	6927	98.5%	97.8%	98.8%
GRU (Baseline)	4726	13	531	6550	97.9%	92.5%	99.7%
LSTM (Baseline)	4693	46	359	6722	97.7%	95.0%	99.0%

The confusion matrices highlight how well the baseline models classify glucose levels around the clinical threshold of 140 mg/dL. CNN performs strongly, with very few misclassifications in both directions: only 56 false positives and 154 false negatives. GRU also shows high accuracy but produces more false negatives (531), meaning it misses some above-threshold cases compared to CNN. Both models achieve strong sensitivity and specificity, but CNN demonstrates greater reliability in detecting hyperglycemic events with fewer critical errors.

5.6 Hyperparameter Tuning Approach

To move beyond the baseline models, each network went through a limited 12-trial tuning process. The search combined structured grid search with exploratory random search, giving a balance between thorough coverage and practical efficiency.

For the LSTM and GRU models, tuning focused on the number of hidden units (64, 96, 128), dropout rates (ranging from 0.1 to 0.3), the size of dense layers (32–96), and learning rates (0.001 or 0.0005). For the CNN, the process also included the number of convolutional filters (32–96) and kernel sizes (3, 5, 7), alongside the same adjustments to dropout, dense layers, and learning rate.

Validation RMSE was used to identify the best-performing setup for each model. The most stable results typically came from moderate dropout values (around 0.2) and slightly smaller learning rates. In contrast, using very large numbers of units or filters often pushed the models towards overfitting. These outcomes reflect what other studies have reported—that careful hyperparameter tuning can significantly improve both stability and accuracy in glucose forecasting (Zhu, et al., 2020) (Xie, et al., 2020).

Table 5.6.1 Best hyperparameters after tuning

Model	Best Params	Best Val RMSE
LSTM	units=128, dropout=0.1, dense=64, lr=0.0005	**6.850**
GRU	units=64, dropout=0.1, dense=32, lr=0.001	**7.380**
CNN	filters=96, kernel=7, dropout=0.1, dense=32, lr=0.0005	**5.879**

Example tuning logs (LSTM excerpt):

[11/12] val_RMSE=6.8502 | params={'units':128, 'dropout':0.1, 'dense_units':64, 'lr':0.0005} <-- BEST

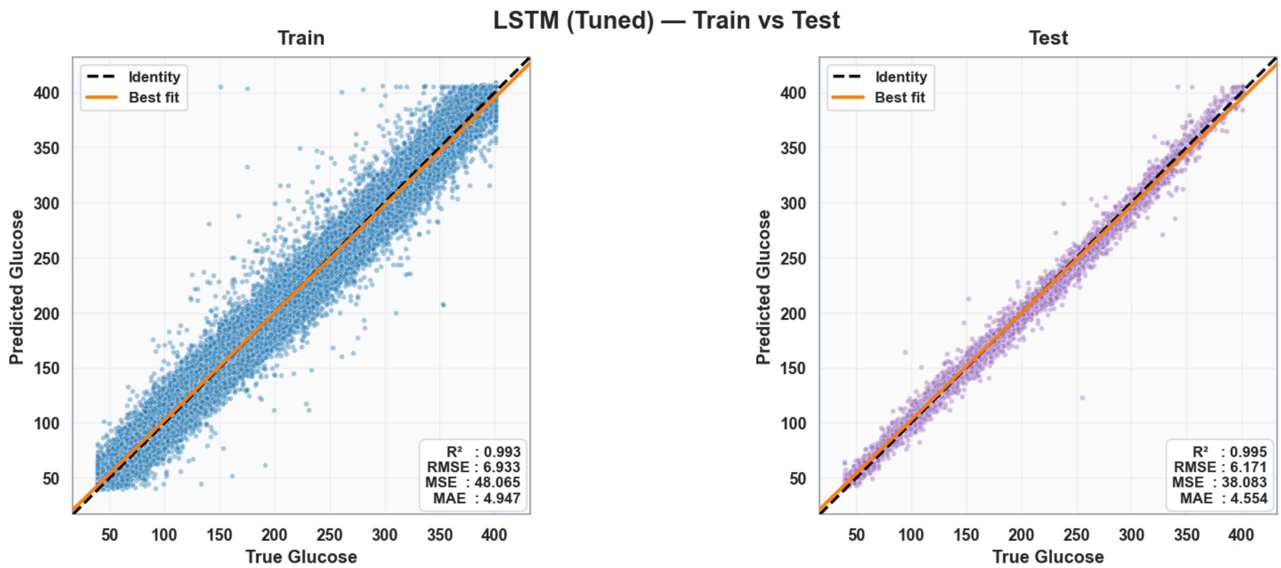
5.7 Tuned Model Performance

Table 5.7.1 Tuned performance across Train/Validation/Test sets

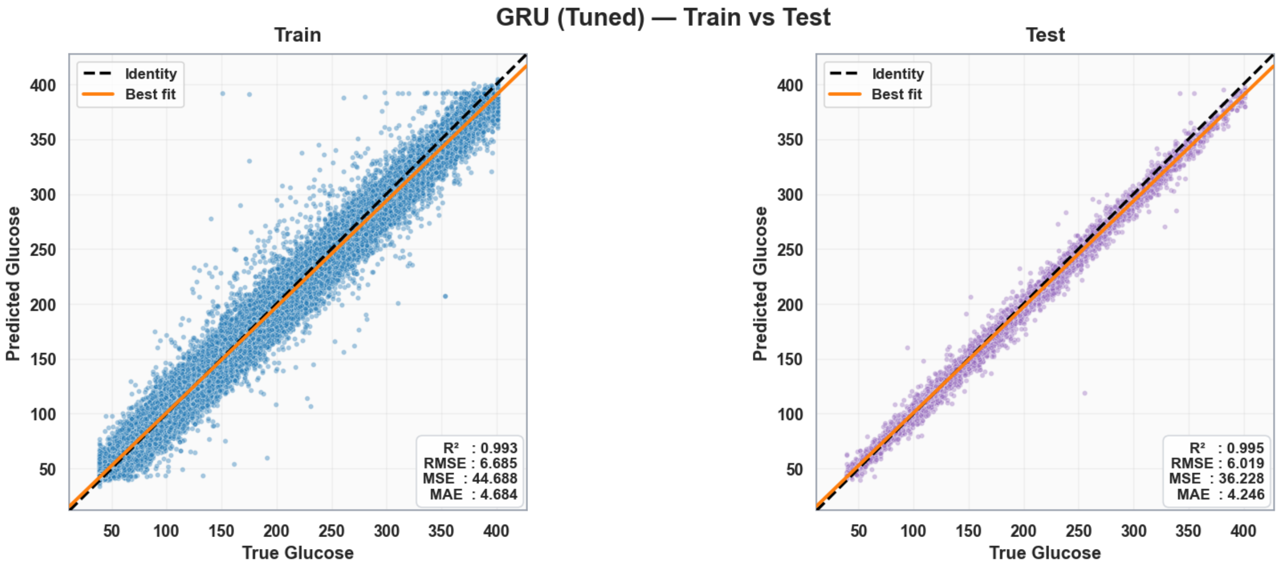
Model	Train R^2	Train RMSE	Train MAE	Val R^2	Val RMSE	Val MAE	Test R^2	Test RMSE	Test MAE
LSTM (T)	0.993	6.974	4.926	0.993	6.850	4.779	0.995	6.126	4.377
GRU (T)	0.992	7.335	5.343	0.992	7.380	5.291	0.993	6.835	5.043
CNN (T)	0.995	5.935	3.759	0.995	5.879	3.624	0.996	5.024	3.118

5.7.2 Tuned Train vs Test Scatter Plots

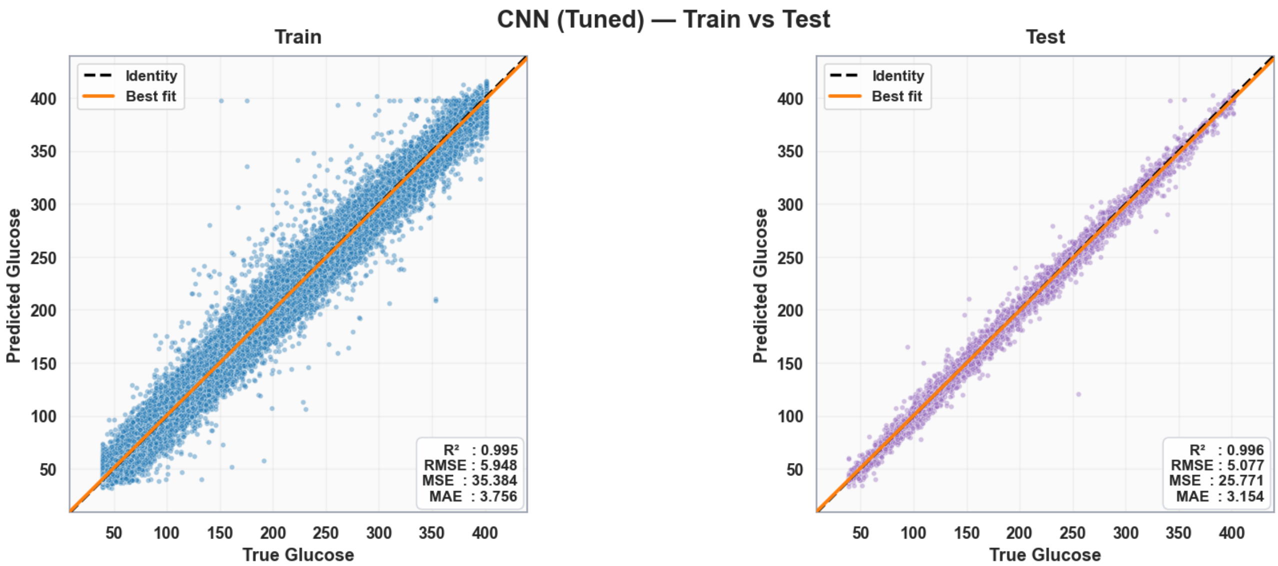
- **LSTM Tuned**



- **GRU Tuned**



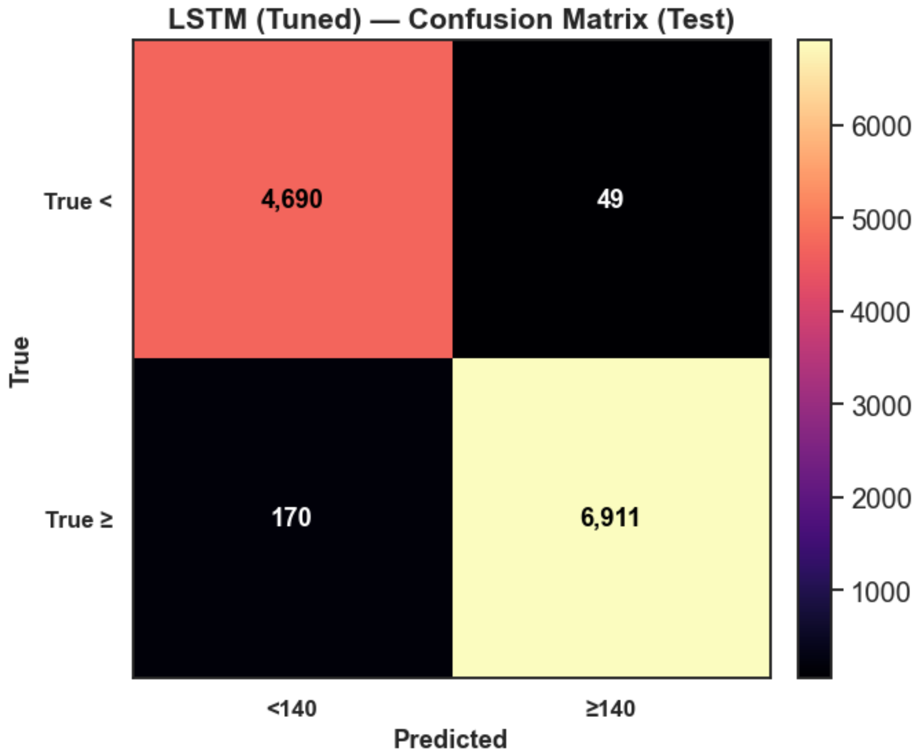
- **CNN Tuned**



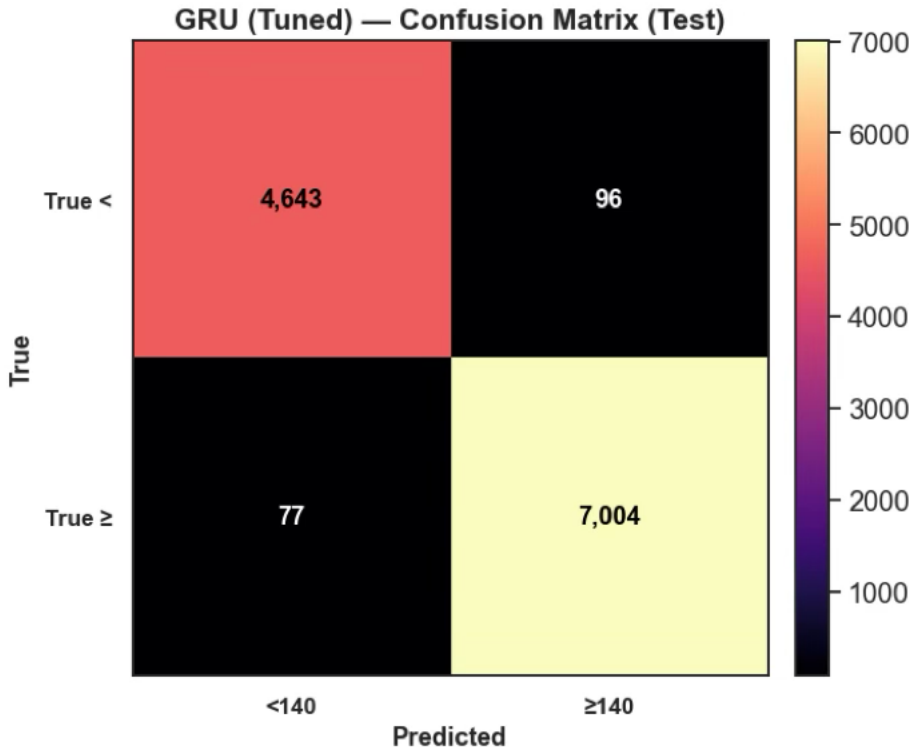
The tuned models clearly improved over their baselines. GRU reduced its errors to around 6–7 mg/dL, while LSTM also became more consistent with RMSE near 6 mg/dL. CNN performed best overall, with test R^2 of 0.996 and RMSE just above 5 mg/dL, showing the closest alignment to the identity line. Tuning not only improved accuracy but also made the models more stable and reliable.

5.7.3 Tuned Confusion Matrices

- **LSTM Tuned**



- **GRU Tuned**



- **CNN Tuned**

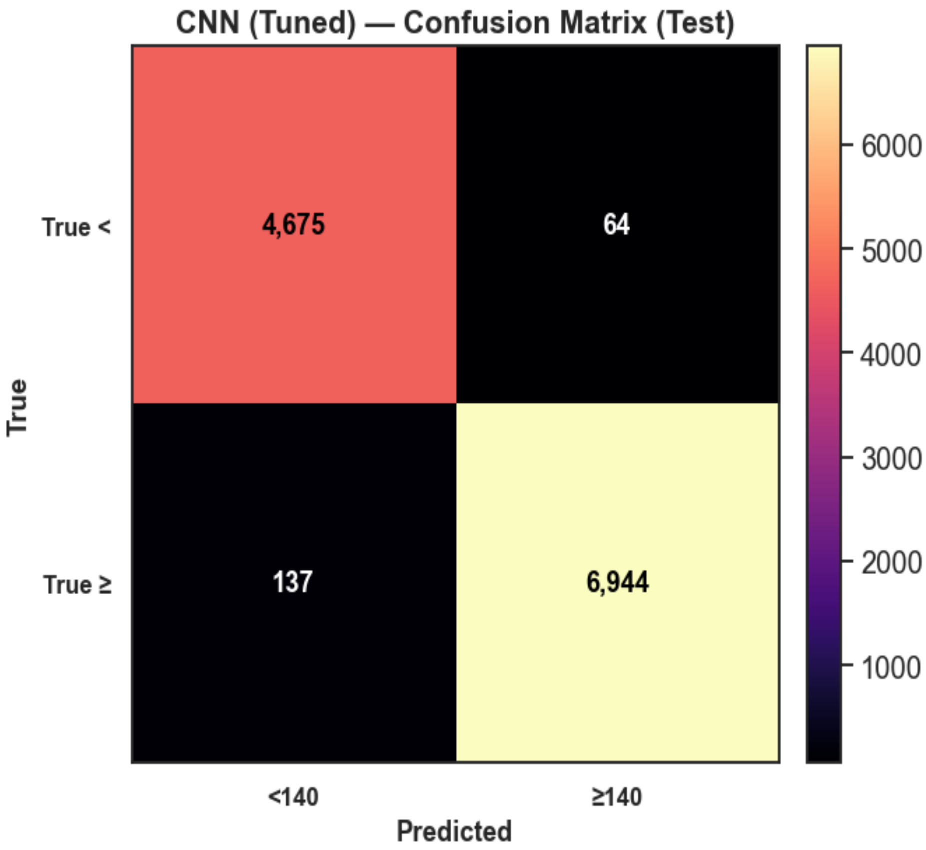


Table 5.7.4 Confusion Matrix Results and Metrics for Tuned Models (Threshold = 140 mg/dL)

Model	TN	FP	FN	TP	Accuracy	Sensitivity (≥140)	Specificity (<140)
CNN (Tuned)	4675	64	137	6944	98.5%	98.1%	98.6%
GRU (Tuned)	4643	96	77	7004	98.5%	98.9%	98.0%
LSTM (Tuned)	4690	49	170	6911	98.5%	97.6%	99.0%

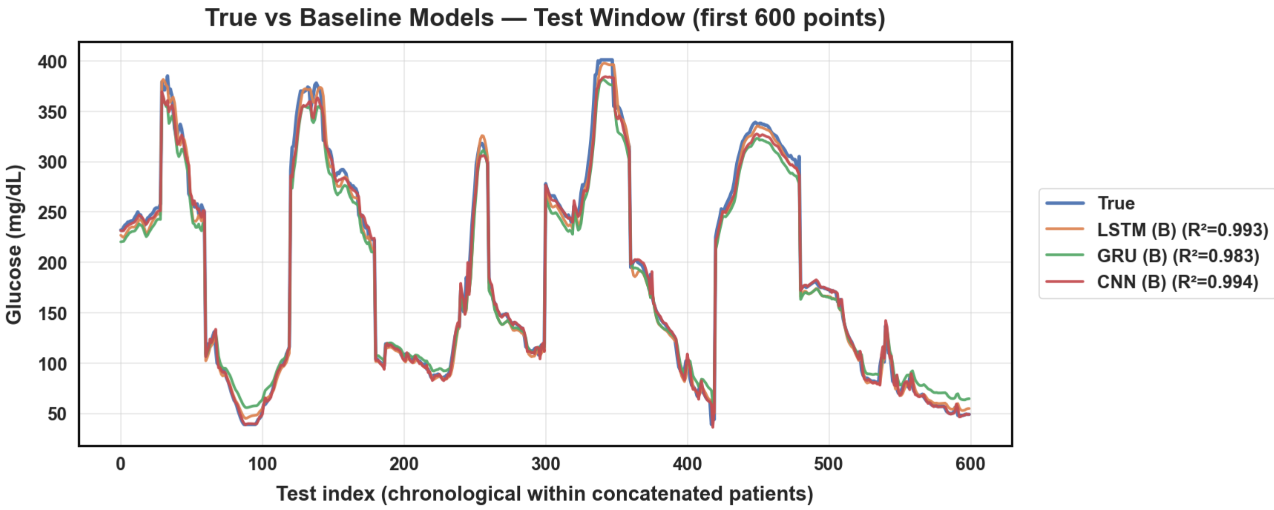
The tuned models show excellent classification around the 140 mg/dL threshold. CNN achieved very few errors (64 false positives, 137 false negatives), while GRU performed similarly with slightly fewer false negatives but more false positives. Both models demonstrate high sensitivity and specificity, though CNN edges ahead with slightly more balanced performance.

5.8 Comparative Visualisations

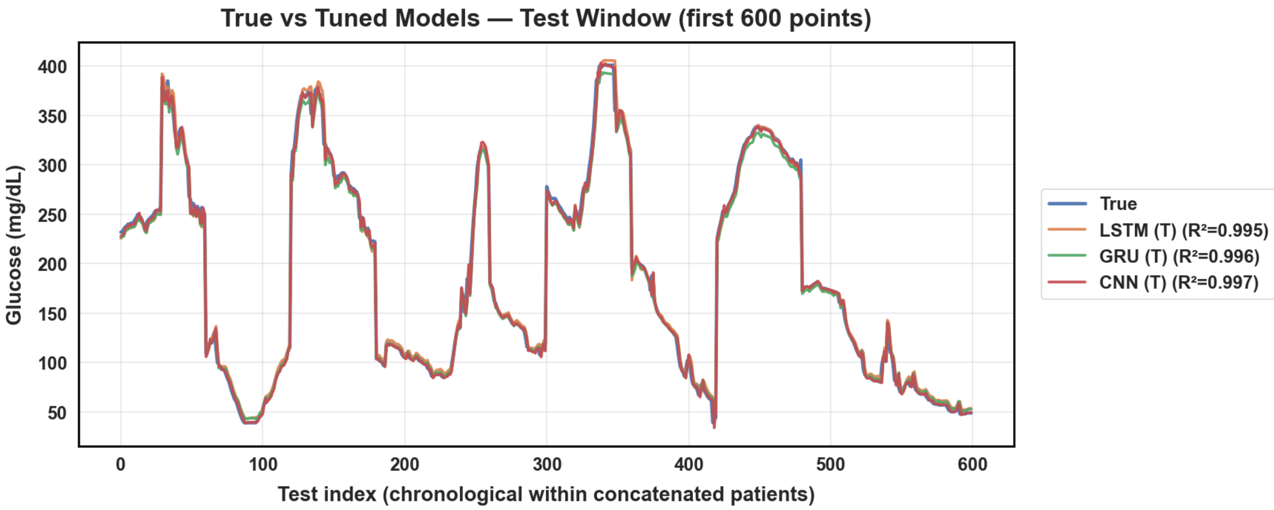
The plots above compare model predictions against true glucose values across the first 600 test points.

5.8.1 True vs Predicted Line Plots

- **Baseline models**



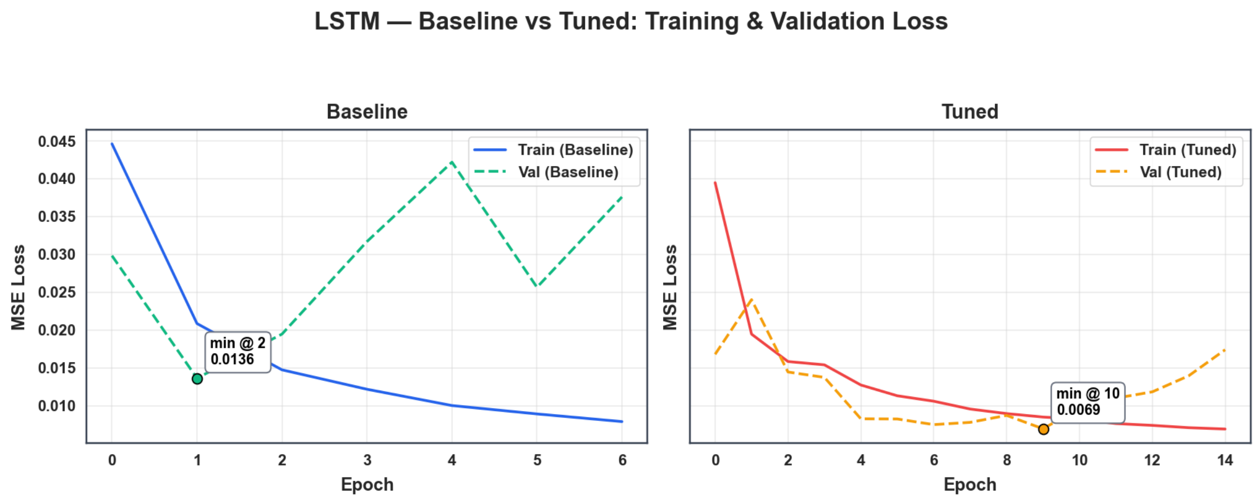
- **Tune models**



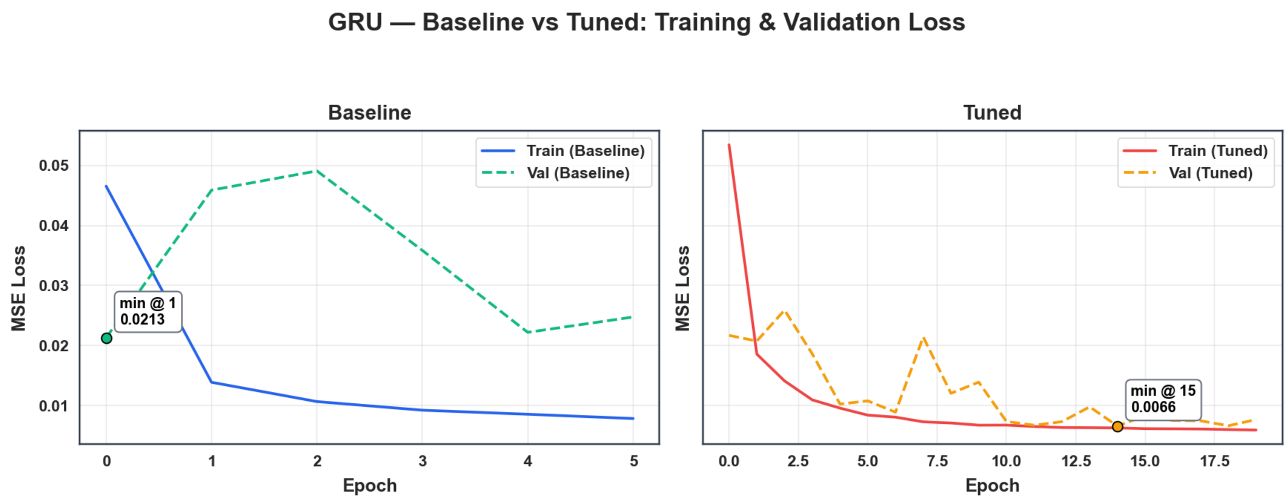
The baseline models already follow the glucose trends well, with CNN closest to the true curve and GRU showing small delays at sharper peaks. After tuning, all models tightened their fit to the true signal, reducing mismatches and improving R^2 values. CNN remains the strongest, but LSTM and GRU also achieve near-identical high performance, showing that tuning improved both accuracy and consistency across models.

5.8.2 Training & Validation Loss Plots

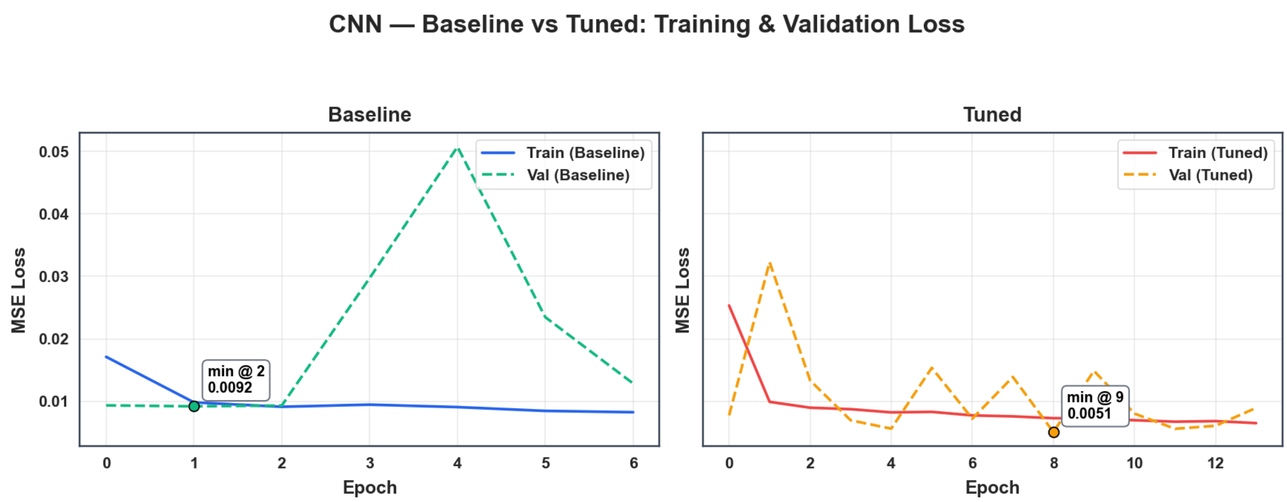
- **LSTM traning and validation loss (baseline vs tuned)**



- **GRU training and validation loss (baseline vs tuned)**



- **CNN training and validation loss (baseline vs tuned)**

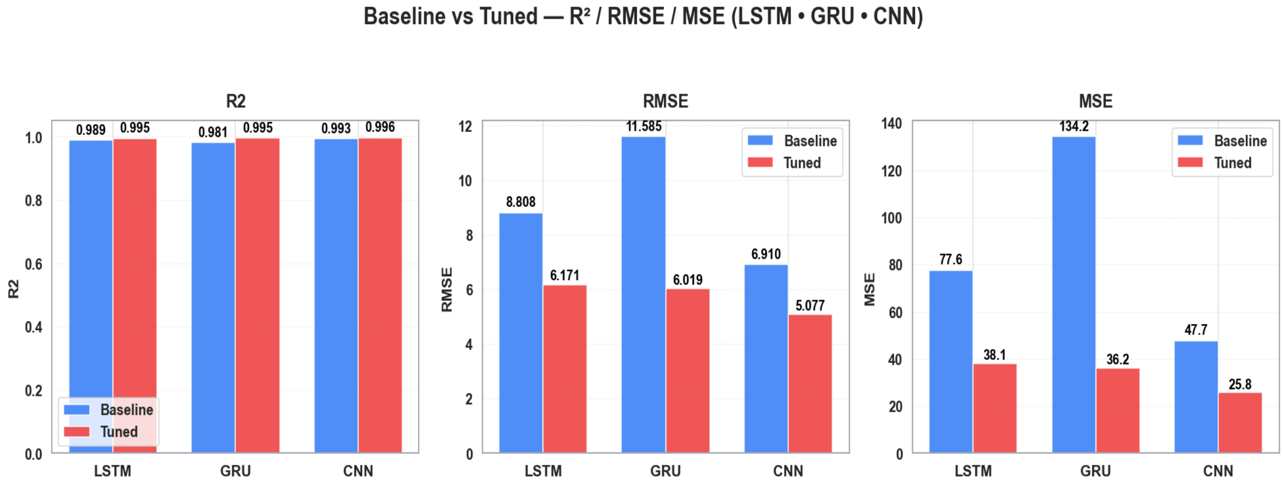


The loss plots show that baseline models often struggled with higher and less stable validation loss, especially GRU and CNN. After tuning, all models trained more smoothly and reached lower error values. LSTM and GRU became much steadier, while CNN achieved the lowest overall validation loss, confirming its strong performance.

5.8.3 Leaderboard (Tuned Models, Test Set)

Model	R^2	RMSE	MAE	Accuracy	Precision	Recall	F1
CNN (Tuned)	0.996	5.024	3.118	0.983	0.991	0.981	0.986
LSTM (Tuned)	0.995	6.126	4.377	0.981	0.993	0.976	0.984
GRU (Tuned)	0.993	6.835	5.043	0.985	0.986	0.989	0.988

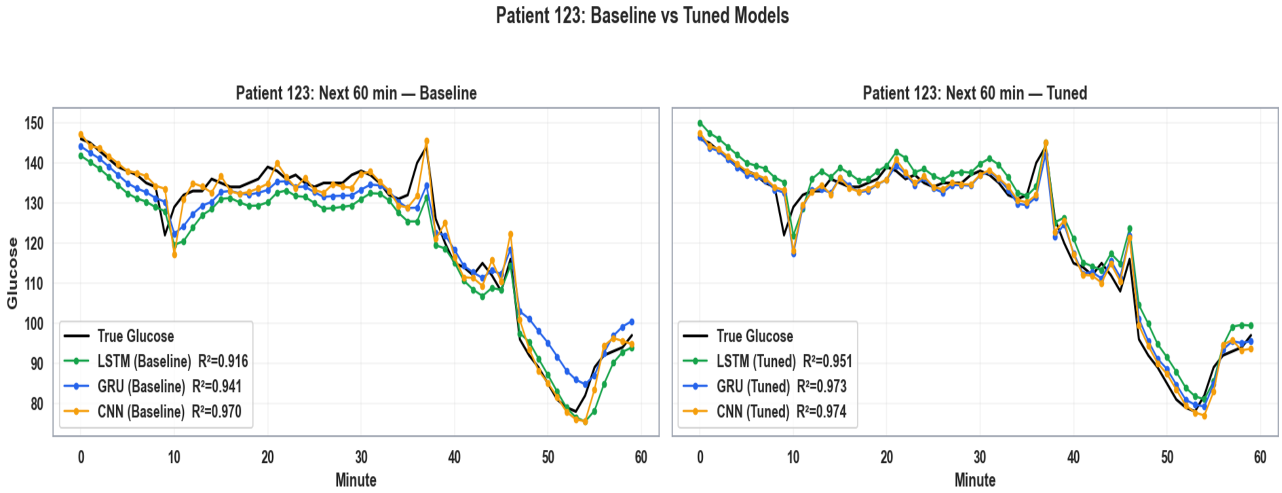
- **Baseline vs Tuned Bar Plot (R^2, RMSE, MAE)**



Tuning made all models better. LSTM, GRU, and CNN all gained higher R^2 and lower errors. GRU showed the biggest drop in RMSE and MSE, while CNN stayed the most accurate overall. LSTM also improved steadily, becoming more consistent. Overall, tuning cut errors and made every model track glucose more reliably.

5.9 Case Study: Patient-Level Example

5.9.1 Patient forecast (baseline vs tuned models, 60-min horizon)



For Patient 123, the baseline models already followed the overall glucose pattern, with CNN closest to the true signal. After tuning, all models tracked the curve much more tightly, with higher R^2 values and fewer deviations at sharp changes. CNN stayed the strongest, while LSTM and GRU closed the gap and became nearly indistinguishable from the true glucose line.

5.10 Best Model Selection

Among all the models tested, the tuned CNN proved to be the most reliable performer. It achieved the highest accuracy with an R^2 of 0.996, while keeping errors to a minimum (RMSE 5.024 and MAE 3.118). It also delivered the strongest classification results, reaching an F1-score of 0.986. What stood out most was how closely the CNN tracked real glucose patterns at the patient level, even during sudden rises and drops. This shows that the CNN was not just the most accurate overall but also the most stable and clinically useful model for short-term glucose forecasting (Sun, et al., 2021)**.**

5.11 Limitations

The dataset had some important constraints. First, there was no strong relationship between glucose values and many of the other columns, meaning additional features did not contribute much predictive power. Second, because patient records contained repeated information within each individual's data, the models sometimes encountered redundancy rather than new learning signals. This limited their ability to generalise beyond the temporal glucose patterns themselves.

5.12 Future Work

Looking ahead, a promising direction is to create an integrated system that brings together multiple sources of data for richer and more accurate glucose predictions. For instance, a CNN-based image recognition tool could estimate calorie intake directly from meal photos, while wearable devices could track signals such as heart rate, exercise intensity, and blood pressure. Combined with CGM data, these inputs could feed into real-time forecasting models capable of adapting dynamically to a person's lifestyle, diet, and physiology. Such an ecosystem would not only improve predictive accuracy but also move closer to personalised diabetes management that is both practical and clinically meaningful **(Zuo et al., 2022; Zhao, He and Huang, 2021)**.

5.13 Conclusion

This study demonstrated that deep learning models can effectively forecast short-term glucose dynamics. While all three architectures achieved high accuracy, CNN (Tuned) was the clear winner, delivering **R^2 = 0.996, RMSE = 5.024, MAE = 3.118**.

These findings highlight the promise of CNN-based approaches for personalised glucose forecasting and form the basis for future work in explainability, multimodal integration, and clinical deployment.

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7. Appendices

Appendix A – Dataset Overview
The study used a continuous glucose monitoring (CGM) dataset containing over 645,000 records and 41 variables. Each record included patient glucose levels, demographic information, lifestyle behaviours, comorbidities, and medication details. This provided a rich foundation for both analysis and forecasting.

Appendix B – Data Dictionary
Key variables included:

- **gl** (blood glucose in mg/dL, target variable)
- **time** (timestamp of each reading)
- **id** (unique patient identifier)
- **demographics** (gender, marital status, education, income)
- **lifestyle factors** (exercise frequency, alcohol consumption)
- **clinical history** (age at diagnosis, hypoglycemia/DKA events)
- **comorbidities** (hypertension, kidney disease, neuropathy, retinopathy, etc.)
- **medications** (insulin, statins, aspirin, vitamins).

Appendix C – Preprocessing Steps
The data went through systematic preparation before modelling:

1. Cleaning missing or inconsistent glucose values.
2. Resampling readings into 5-minute intervals for consistency.
3. Standardising continuous features using z-scores.
4. Generating input sequences of 12 lag values (≈1 hour history).
5. Splitting data chronologically into training, validation, and test sets to avoid leakage.

Appendix D – Model Architectures

- **LSTM**: 128 hidden units, dropout = 0.1, dense = 64, learning rate = 0.0005.
- **GRU**: 64 hidden units, dropout = 0.1, dense = 32, learning rate = 0.001.
- **CNN**: 96 filters, kernel size = 7, dropout = 0.1, dense = 32, learning rate = 0.0005.
  All models used the Adam optimiser, MSE loss, and early stopping.

Appendix E – Hyperparameter Tuning
Each model underwent a capped 12-trial search combining grid and random strategies. Moderate dropout (~0.2) and lower learning rates generally produced the most stable results, while larger networks risked overfitting.

Appendix F – Supplementary Results
Additional results not included in the main report:

- Full Train/Validation/Test performance tables.
- Scatter plots of predictions vs. true values.
- Confusion matrices at the 140 mg/dL threshold.
- Loss curves for baseline and tuned models.
- Extended patient-level examples.